

CE 310 — Week 4 Pre-Lab Quiz

Week: 4 | Topic: Discrete Events: The Poisson Distribution Points: 10 questions × 1 pt each = 10 pts Submission: Complete in D2L before class begins. No partial credit for MC.

Instructions

Select the single best answer for each question. The quiz covers the Poisson distribution, event rates, PMF vs. CDF, return periods, and the Excel functions you will use in today's lab. Review the Week 4 Background reading before completing this quiz.

Questions

Q1. In October 2015, a storm was labeled a “1,000-year flood” — yet similar events had occurred in the same city twice in the previous six years. The Poisson model reveals that:

- (A) The three events confirm that 1,000-year floods are becoming more frequent due to climate change — no statistical error was made.
 - (B) Three events in six years is consistent with a 1,000-year return period if the storms were clustered.
 - (C) $P(X \geq 3 \text{ in 6 years} \mid \lambda = 0.006)$ is essentially zero, which means the “1,000-year” label was almost certainly incorrect.
 - (D) The Poisson model is inappropriate for flood events because rainfall is not discrete.
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Q2. A civil engineer says: “The 50-year storm will not happen during this 10-year project.” Why is this reasoning dangerous?

- (A) Because the 50-year storm has already occurred within the last decade, resetting the clock.
 - (B) Because $P(\text{at least one 50-year storm in 10 years}) = 1 - e^{-(10/50)} \approx 18\%$, which is a meaningful probability that engineers cannot ignore.
 - (C) Because ASCE 7-22 requires designing for the 100-year storm, not the 50-year storm.
 - (D) Because the engineer is confusing the return period with the recurrence interval.
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Q3. The Poisson distribution models which type of situation most accurately?

- (A) The height of concrete cylinder test samples — a continuous measurement.
- (B) The number of equipment failures in a building per year, given a known average failure rate.
- (C) The probability that a single component either passes or fails an inspection.

(D) The total energy consumption of a building over a month.

Q4. The dataset for this week's lab covers 10 years of daily precipitation at Tucson International Airport. How many rows does the dataset contain?

- (A) 365
 - (B) 1,200
 - (C) 3,650
 - (D) 3,652
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Q5. Tucson's North American Monsoon produces most of the city's annual rainfall in July and August. The dataset records 103 rain days in July across 10 years. What is the correct July lambda (λ_{july})?

- (A) $103 / 120 = 0.858$ rain days per month
 - (B) $103 / 31 = 3.32$ rain days per July day
 - (C) $103 / 10 = 10.3$ rain days per July
 - (D) $103 / 30 = 3.43$ rain days per summer month
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Q6. Which statement correctly distinguishes the Poisson PMF from the Poisson CDF?

- (A) The PMF gives the probability of at least k events; the CDF gives the probability of exactly k events.
 - (B) The PMF gives the probability of exactly k events; the CDF gives the probability of k or fewer events.
 - (C) The PMF is used for continuous variables; the CDF is used for discrete count variables.
 - (D) The PMF and CDF are identical when lambda equals 1.
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Q7. For a Poisson random variable X with $\lambda = 3.86$, what is $E[X]$ (the expected value)?

- (A) $\sqrt{3.86} \approx 1.96$
 - (B) 3.86
 - (C) $3.86^2 = 14.90$
 - (D) Cannot be determined without the full PMF table.
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Q8. A facilities team estimates $\lambda = 2$ equipment faults per quarter. Using the Poisson PMF formula by hand (not Excel) and $e^{-2} \approx 0.1353$, what is $P(X = 1)$?

- (A) 0.1353 — this is $P(X = 0)$, not $P(X = 1)$
- (B) 0.2706 — $P(X = 1) = e^{-2} \times 2^1 / 1! = 0.1353 \times 2$
- (C) 0.0677 — dividing by $1!$ as if it were $2!$

(D) $0.4055 = e^{-2} \times 3$, an arithmetic slip

Q9. In Excel, you want to compute $P(X = 7)$ for a Poisson distribution with $\lambda = 10.3$. Which formula is correct?

- (A) =POISSON.DIST(10.3, 7, FALSE)
 - (B) =POISSON.DIST(7, 10.3, TRUE)
 - (C) =POISSON.DIST(7, 10.3, FALSE)
 - (D) =NORM.DIST(7, 10.3, SQRT(10.3), TRUE)
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Q10. You want to compute $P(X > 5)$ for a Poisson distribution with $\lambda = 3.86$. Which Excel formula is correct?

- (A) =POISSON.DIST(5, 3.86, FALSE)
 - (B) =1 - POISSON.DIST(5, 3.86, FALSE)
 - (C) =1 - POISSON.DIST(5, 3.86, TRUE)
 - (D) =POISSON.DIST(6, 3.86, TRUE)
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